



VICTORIA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATION 2018

H2 Mathematics

9758/01

Paper 1

3 hours

Additional Materials: Answer Paper
Graph Paper (Upon Request)
List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

- 1 Abel, Mo and Paula train together for running. In a particular year, they spent time on three types of training programmes called “Circuit”, “Intervals” and “Long Run”.

For each type of training programme, the duration of each training session is the same for all three runners, except that Mo requires 10% more training time for each “Interval” session compared to Abel and Paula.

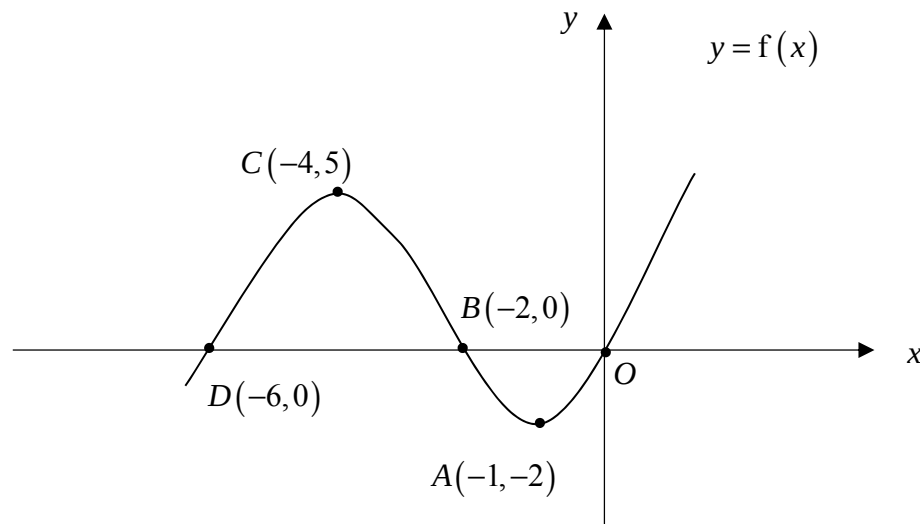
The table below shows the number of training sessions each runner attended, as well as their total training time for the year.

	Number of training sessions in the year			Total training time (in minutes)
	“Circuit”	“Intervals”	“Long Run”	
Abel	23	40	61	12600
Mo	34	67	75	17600.5
Paula	33	53	87	17725

Find the total amount of time (in minutes) each runner spent on “Intervals” in the year.

[4]

2

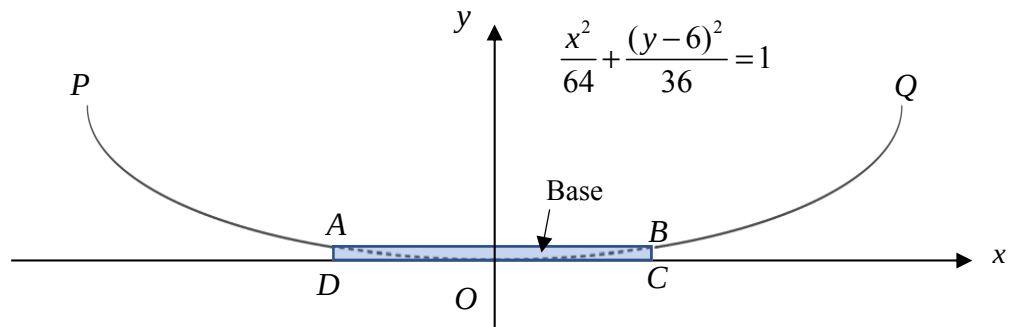


The diagram shows the graph of $y = f(x)$. The curve passes through the points $A(-1, -2)$, $B(-2, 0)$, $C(-4, 5)$, $D(-6, 0)$ and the origin O . Points A and C are the minimum point and maximum point respectively. On separate diagrams, sketch each of the following graphs indicating where possible, the coordinates of the stationary points, the points of intersections with the axes and the equations of any asymptotes.

(i) $y = f\left(\frac{x}{2} + 2\right)$, [2]

(ii) $y = \frac{1}{f(x)}$. [3]

3



The diagram above shows the cross section of a bowl $PABQ$ and a base $ABCD$ of height h units. The curve $\frac{x^2}{64} + \frac{(y-6)^2}{36} = 1$, $h \leq y \leq 6$, from point P to A and from point B to Q is rotated through π radians about the y -axis to form the curved surface of the bowl, while the rectangle $ABCD$ is rotated through π radians about the y -axis to form the solid cylindrical base.

The bowl is assumed to have negligible thickness. Determine, correct to 3 decimal places, the greatest height of the base such that volume of the bowl is at least 790 units³. [5]

4 A curve C has parametric equations

$$x = a \cos(2t), \quad y = 2a \tan t,$$

where $0 \leq t < \frac{\pi}{2}$ and a is a positive constant.

- (i) Sketch C , showing clearly the coordinates of any point(s) of intersection with the axes and the equations of any asymptotes. [3]
- (ii) It is given that $a = 3$. Find the equation of the tangent to C at the point where $t = \frac{\pi}{3}$. [3]

5 Using an algebraic method, solve the inequality $\frac{7x+6}{x+2} \leq 3x-1$. [4]

Hence, find the set of values of x that satisfy $\frac{7e^{-x}+6}{e^{-x}+2} \leq 3e^{-x}-1$. [2]

6 Do not use a calculator in answering this question.

- (a) Express $\left(\frac{\sqrt{2} + i\sqrt{2}}{1 - i\sqrt{3}}\right)^8$ in the cartesian form $a + ib$, where a and b are in a non-trigonometrical form. [3]

- (b) Solve the simultaneous equations

$$2iz^* - w = 1 + 7i \text{ and } z + 2w = 16 - 2i,$$

giving z and w in the cartesian form $p + iq$. [4]

- 7 Given that $y = \tan^{-1}(ax + 1)$, where a is a constant, show that $\frac{dy}{dx} = a \cos^2 y$. Use this result to find the Maclaurin series for y in terms of a , up to and including the term in x^3 . [6]

Given that the first two terms in the series expansion of $\sqrt{9 - x}$ are equal to the first two terms in the series expansion of $b \tan^{-1}(ax + 1)$, find the exact values of a and b . [3]

- 8 (a) The straight line l has equation $\mathbf{r} = \mathbf{i} + 6\mathbf{j} - 3\mathbf{k} + s(\mathbf{i} - 2\mathbf{j} + 2\mathbf{k})$, $s \in \mathbb{R}$. The plane p has equation $\mathbf{r} \cdot (2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}) = 6$. The line l intersects the plane p at a point C .

(i) Show that the position vector of C is $3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$. [2]

(ii) Find a vector equation of the line which lies in p , passes through C and is perpendicular to l . [3]

- (b) Referred to the origin O , the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively.

(i) Show that $|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2$. [3]

It is further given that $\mathbf{a}$ is a unit vector, $\mathbf{b}$ has magnitude 3 and $\mathbf{a} \cdot \mathbf{b} = \frac{1}{3}$.

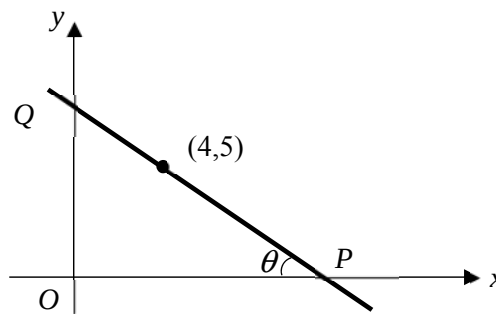
(ii) Find the exact area of triangle OAB . [2]

(iii) Give the geometrical meaning of $|\mathbf{a} \cdot \mathbf{b}|$. [1]

9 The curve C has equation $y = \frac{2x+5}{4-x^2}$.

- (i) Sketch C , giving the coordinates of the axial intercepts, turning points and equations of any asymptotes. [4]
- (ii) Find the numerical value of the area bounded by C and the line $y = 2$. [3]
- (iii) Find the set of values of k such that $\left| \frac{2x+5}{x^2-4} \right| = k$ has 3 distinct negative real solutions. [3]

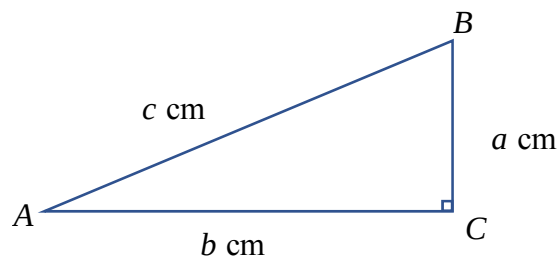
10 (a)



A line passes through the point $(4, 5)$ and cuts the x -axis and y -axis at points P and Q respectively. It is given that angle $OPQ = \theta$, where $0 < \theta < \frac{\pi}{2}$ and O is the origin. You may assume that the same scale is used on both axes.

- (i) Show that the equation of line PQ is given by $y = (4 - x) \tan \theta + 5$. [3]
- (ii) Hence or otherwise, show that $OP + OQ = 9 + 4 \tan \theta + 5 \cot \theta$. By differentiation, find the stationary value of $OP + OQ$ as θ varies. Determine the nature of this stationary value. [5]

(b)

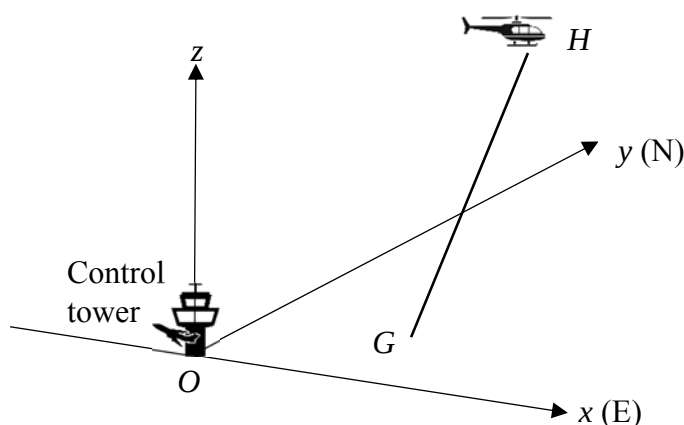


The diagram above shows a right angled triangle ABC . The lengths AB , BC and AC are denoted by c cm, a cm and b cm respectively. The triangle has a fixed area of 100 cm^2 .

Express b in terms of a . [1]

Given that BC increases at a rate of 3 cm s^{-1} , find the rate of change of c when $a = 20$. [3]

11



The diagram illustrates the initial flight path of a helicopter H taking off from an airport. The origin O is taken to be at the base of the control tower. The x -axis is due east, the y -axis due north, and the z -axis is vertical. The units of distances are measured in kilometres.

The helicopter takes off from the point G on the ground. The position vector $\mathbf{r}$ of the helicopter t minutes after take-off is given by

$$\mathbf{r} = (1 + t)\mathbf{i} + (0.5 + 2t)\mathbf{j} + 2t\mathbf{k}.$$

- (i) Write down the coordinates of G and describe the initial flight path. [3]
- (ii) Find the acute angle that the helicopter's flight path makes with the horizontal. [3]
- (iii) A mountain top is situated at the point $M(5, 4.5, 3)$. Determine how long after take off the helicopter will be nearest to M . [2]
- (iv) An eagle sets off from the mountain top to hunt for food. The position of the eagle satisfies the equation

$$\frac{x-2}{3} = \frac{z-2}{1}, y = 4.5.$$

Determine if the flight path of the helicopter will intersect the path traced out by the eagle, showing your reasoning clearly. [3]

- (v) The helicopter enters a cloud at a height of 2 km. Given that the visibility on that day is 3.75 km, determine if the air traffic controller who is situated at 70 m above ground level, in the control tower, will be able to sight the helicopter as it enters the cloud. [2]

- 12** The logistic equation, sometimes called the *Verhulst model*, is a model of population growth. Letting N be the population size at any time, t (in years), this model is formalized by the differential equation:

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right),$$

where r and K are real positive constants.

- (i) By considering the value of $\frac{dN}{dt}$ when N approaches K , explain, in the context of the question, the significance of K . [2]
- (ii) Solve the differential equation, and show that $N = \frac{K}{1 + Be^{-rt}}$, where B is a constant. [5]
- (iii) It is now given that the initial population in a small town is 10,000. The rate of population growth at that time is 100 per year. When the population is 15,000, the population growth rate is 75 per year. Use the *Verhulst model* to find N in terms of t .
Hence, sketch the graph of N against t . [5]

[End of Paper]